



Code	Subject Title	Cr. Hrs	Semester
STAT-312	Advanced Sampling Techniques (Theory)	3	VI
Year	Discipline		
3	Statistics-I,II,III		

Code	Subject Title	Cr. Hrs	Semester
STAT-313	Advanced Sampling Techniques (Lab)	1	VI
Year	Discipline		
3	Statistics-I,II,III		

Advanced Sampling Techniques

Course Outline

1. Ratio and Regression estimation. Estimation of total, mean square error and bias using design based approach and model based approach in estimate in simple random sampling. Unbiased ratio-type estimators. Ratio estimation in stratified sampling. Estimation of mean and variance in linear regression estimates. Best linear unbiased estimator (BLUE). Bias of the linear regression estimates. Regression estimation in stratified sampling. The Linear regression estimator under the general linear model.
2. Cluster sampling. Estimation of mean, total and variance for single-stage cluster sampling. Cost function. Variance function, cluster sampling for proportions, Sampling with unequal probability with replacement.
3. Two-stage sampling. Estimation of mean, total, proportion and variance. Both stages with equal probability. Two-stage sampling with units of unequal sizes, First stage PPS (with replacement) and second stage with equal probability. Both stages with probability proportional to size and with replacement. Sampling methods when a single primary unit is selected for the sample. Basic concept of double sampling.

Books Recommended

1. Cochran, W.G. "Sampling Techniques" John Wiley & Sons, New York, Third Edition, 1977.
2. Raj, D., "Sampling Theory" Mc-Graw-Hill Book Company, New York, 1971.
3. Singh, D. Chaudhry F.S. "Theory and Analysis of Sample Survey Designs", Wiley Eastern Limited, New Dehli, India, 1986.

Reference Books

1. Fuller, Wayne A. "Sampling Statistics" John Wiley and Sons, New Jersey, 2009.
2. Brewer, K. "Combined Survey Sampling Inference" Oxford University Press, New York, 2002.
3. Raj, D. "Design of Sample Survey" Mc-Graw-Hill Book Company, New York, 1971.
4. Kish, L. "Survey Sampling" John Wiley and Sons, New York, 1965.
5. Som, R.K. "A Manual of Sampling Techniques" Heinemaan Educational Books Limited, London, 1973.
6. Sukhatme, P.V. and Sukhatme, B.V., Sukhatme, S. and Asok, C. "Sampling Theory of Surveys with Applications" Iowa State University Press, Ames, IOWA. Third Edition, 1984.

Course Title:	Advanced Computer Languages
Course Code:	STAT-319
Semester:	VI
Credit Hours:	3 Credit Hours
Pre-requisites:	N / A

Learning Outcomes

By the end of this course, students will be able to:

1. Get familiar with the interface of R along with the objects required for the purpose of data analysis.
2. Learn the basic programming skills including logical statements, looping and graphical functions.
3. Generate random numbers and simulate data from different distributions.
4. Estimate Regression and Time Series models based on Monte Carlo Simulations and Bootstrapping.

Course Outline

Unit 1

1.1 Introduction to R and its Framework

Downloading, Installing and Starting R and associate libraries. Calculating environment of R, Types of R objects, Vector, Matrix, Data frame, Array etc. Writing Scripts, Basic programming skills, Logical statements, Looping, Programming flow and basic debugging. Using built-in functions, Input and Output files, Programming with functions, Graphics.

Unit 2

2.1 Empirical Study of Sampling Distributions

Probability and probability distributions, Generating random numbers, Selecting random samples, Empirical study of the sampling distribution of estimators.

2.2 Data Simulation

Simulation of data from a probability distribution, Simulation of data for a regression model, Simulation of data for time series model, Monte Carlo simulation, Bootstrapping.

Text Book

1. Lander, J.P. (2017). R for everyone: Advanced Analytics and Graphics, 2nd edition. Pearson Education Inc. New York.
2. Purohit, S. G., Gore, S. G., & Deshmukh, S. R. (2008). *Statistics Using R*. Narosa Publishing House

Suggested Readings:

1. Fischetti, A. (2018). *Data Analysis with R: A comprehensive guide to manipulating, analyzing and visualizing data in R*. Packt Publishing Ltd.
2. Jones, O., Maillardet, R., & Robinson, A. (2014). *Introduction to scientific programming and simulation using R*. Chapman and Hall/CRC.



Code	Subject Title	Cr. Hrs	Semester
STAT-310	Advanced Experimental Design (Theory)	3	VI
Year	Discipline		
3	Statistics-I,II,III		

Code	Subject Title	Cr. Hrs	Semester
STAT-311	Advanced Experimental Design (Lab)	1	VI
Year	Discipline		
3	Statistics-I,II,III		

Course Outline

1. Factorial experiments and its advantages. 1st series Factorial in Randomized Complete Block designs. 2nd series Factorial experiments. Linear and quadratic components of main effects and interactions. 3rd series Factorial experiments.
2. Confounding, its types and its advantages. Complete and partial confounding in 2nd series.
3. Fractional replication. Quasi-Latin squares.
4. Split-plot designs and Split-split plot designs.
5. Balanced incomplete and Partially Balanced incomplete block designs. Comparison of Incomplete Block design with Randomized Complete Block design. Youden Squares.

Books Recommended

1. Cochran, W.C. and Cox, G.M. "Experimental Design", John Wiley and Sons, New York, Second Edition, 1957.
2. Montgomery, D.C. "The Design and Analysis of Experiments", John Wiley and Sons, New York, Fourth Edition, 1997.
3. John, J.A. and Quenouille, M.H. "Experiments Design and Analysis", Second Edition, Charles Griffin & Co. London, 1977.

Reference Books

1. Kempthorne, O. & Hinkelmann, K. "Design and Analysis of Experiments, Vol.1", John Wiley and Sons, New York, 1994.
2. Barker, T.B. "Quality by Exp. Design", Marcel Dekker, Inc. New York, Second Edition, 1994.
3. Boniface, D.R., "Experiment Design and Statistical Methods for Behavioural and Social Research", Chapman & Hall, London. First Edition, 1995.
4. Ostle, B. and Menging, R.W. "Statistics in Research", The Iowa State University Press, Third Edition, 1975.
5. Winer, B.J. "Statistical Principles in Experimental Design". McGraw-Hill Book Company, New York, Second Edition, 1971.
6. Federer, W.T. "Experimental Design", Macmillan Company, New York, 1955.
7. Graybill, F.A. "An Introduction to Linear Statistical Models, Vol.1" McGraw Hill Book Company, New York, 1961.
8. Heath, D. "An Introduction to Experimental Design and Statistics for Biology", UCI Press, London, second edition, 1996.



9. Clewer, AlanG, "Practical Statistics and Experimental Design for Plant and Crop Science", Wiley N.Y., 2001.
10. Quinn Gerry P, "Exp. Design and Data Analysis for Biologists" Camb. Press, Cambridge, 2002.
11. Jeff Wu, C.F. "Experimental: Planning Analysis", Wiley, New York, 2002.
12. Kuehl, R.O. "Design of experiments: Statistical principles of research design and analysis" Duxbury, Boston, 2000.

Code	Subject Title	Cr. Hrs	Semester
STAT-314	Multivariate Techniques (Theory)	3	VI
Year	Discipline		
3	Statistics-I,II,III		

Code	Subject Title	Cr. Hrs	Semester
STAT-315	Multivariate Techniques (Lab)	1	VI
Year	Discipline		
3	Statistics-I,II,III		

Multivariate Techniques

Course Outline:

Review of matrix algebra, Notions of multivariate distributions. The multivariate normal distribution and its properties. Linear compound and linear combinations. Estimation of the mean vector and the covariance matrix. The Wishart distribution and its properties. The joint distribution of the sample mean vector and the sample covariance matrix.

Books Recommended

1. Johnson, R.A., & Wichern, D.W. (2008). Applied multivariate statistical analysis, Pearson Education: Singapore.
2. Anderson, T.W. (2003). An introduction to multivariate statistical analysis (3rd ed.). John Wiley & Sons: New York.
3. Chatfield, C. & Collins, A.J. (1980). Introduction to multivariate analysis. Chapman and Hall: London.
4. Morrison, D.F. (1990). Multivariate statistical methods (3rd ed.). McGraw Hill Publishing Co.: New York.

Reference Books

1. Kandall, M.G., & Stuart, A. (1983). The advanced theory of statistics (4th ed.). Charles Griffin and Company: London.
2. Rao, C.R. (1973). Linear statistical inference and its applications (2nd ed.). John Wiley and Sons: New York.



Code	Subject Title	Cr. Hrs	Semester
STAT-316	Probability Distributions (Theory)	3	VI
Year	Discipline		
3	Statistics-I,II,III		

Code	Subject Title	Cr. Hrs	Semester
STAT-317	Probability Distributions (Lab)	1	VI
Year	Discipline		
3	Statistics-I,II,III		

Probability Distributions

Course Outline

Continuous distributions: continuous uniform, normal, exponential, gamma, beta, lognormal, Weibull, Pareto and Cauchy distributions and their properties.

Bivariate distributions. Marginal distribution. Conditional distribution and independence. Conditional expectation and conditional variance. Bivariate normal distribution and its properties.

Transformation of random variables. Sum, product and quotient of random variables. Moment generating function techniques. Derivations of chi-square, t and F distributions and their properties. Order statistics, Distribution of the rth order statistics. Distribution of sample range, sample median and sample mid-range.

Books Recommended

1. Hogg, R.V. and Craig, A.T., "Introduction to Mathematical Statistics" Prentice-Hall International, Inc. Engle Wood Cliffs, N.T., Fifth Edition, 1995.
2. Mood, A.M., Graybill, F.A. and Boes, D.C. "Introduction to the Theory of Statistics" McGraw-Hill Book Company, New York, Third Edition, 1974.

Reference Books

1. J. Susan Milton and Jesse C. Arnold, "Introduction to probability and statistics", McGraw Hill, 2003.
2. Sheldon, M. Ross, "Introduction to probability modes", Academic press, 2003.
3. Dudewicz, E.J. and Misra, S.N. "Modern Mathematical Statistics" John Wiley and Sons, New York, 1988.
4. Hogg, R.V. and Tanis, E.A. "Probability and Statistical Inference" McMillan Publishing Company, New York, Forth Edition, 1993.
5. Stuart, A. and Ord, J.K. "Kendall's Advanced Theory of Statistics Vol.-I" Edward Arnold, London, Sixth Edition, 1994.